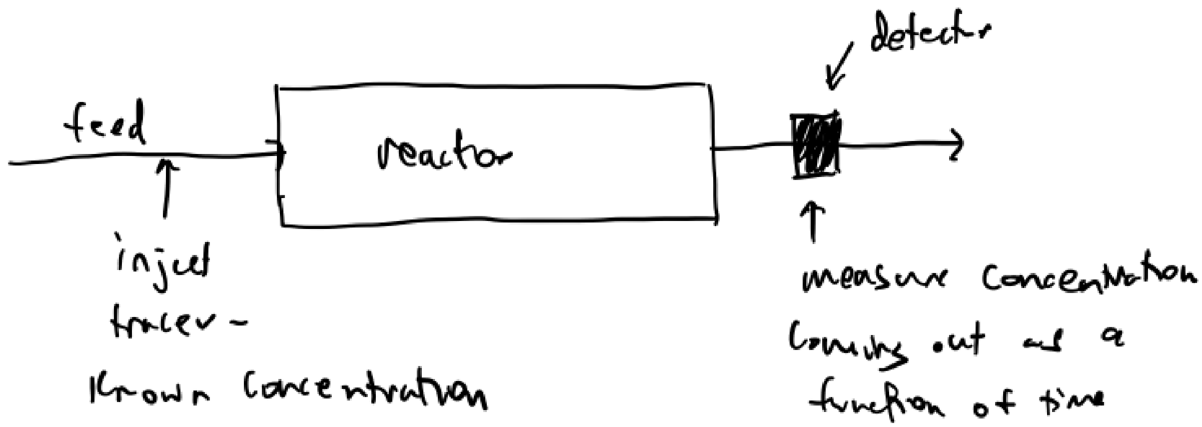
26 Pulse and step experiments

Residence time distributions (RTDs) are measured by dosing a "tracer" to the reactor.



A tracer

- Is non- reactive
- Is easily detectable

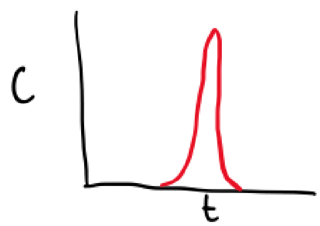
- Has physical properties similar to the reactant mixture & is completely soluble in the reactant/product mixture
- Is often a colored material, a radioactive material, or an inert gas

The tracer can be dosed in either a pulse or step experiment.

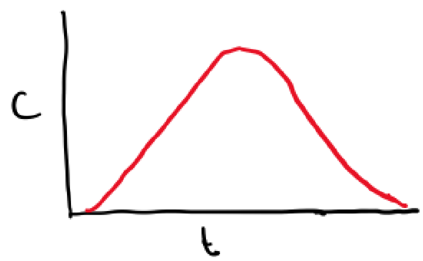
Pulse experiment: An amount of tracer is suddenly injected in one shot

The time needed to collect all of the tracer is used to determine the RTD.

at reactor inlet

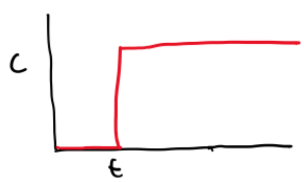


at reactor outlet

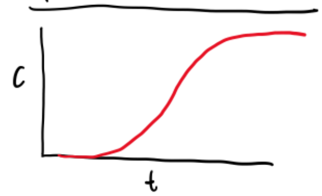


Step experiment: A concentration of tracer is constantly fed to the reactor

at the reactor inlet



at the reactor outlet



$$C_{in}(t) = \begin{cases} 0 & t < 0 \\ C_0 & t \geq 0 \end{cases}$$

The time needed for the concentration of tracer exiting the reactor to equal that entering the reactor is used to determine the RTD.

Important RTD statistics:

- ΔN: Amount of tracer leaving the reactor between t and Δt:

$$\Delta N = C(t) v \Delta t$$

from the $C(t)$ curve @ the reactor outlet

- $\frac{\Delta N}{N_0}$: The fraction of tracer/reactor contents that has a residence time between t and Δt:

$$\frac{\Delta N}{N_0} = \frac{v C(t) \Delta t}{N_0}$$

, N_0 is the total amount that was injected

$$N_0 = \int_0^\infty v C(t) dt$$

- $E(t)$: Residence time distribution

$$E(t) = \frac{\frac{\Delta N}{N_0}}{\Delta t} = \frac{v C(t) \Delta t}{N_0 \Delta t} = \frac{v C(t)}{N_0} = \frac{v C(t)}{\int_0^\infty v C(t) dt}$$

if v is constant, $E(t) = \frac{C(t)}{\int_0^\infty C(t) dt}$

The fraction of tracer/reactor contents leaving the reactor that has resided in the reactor for a time between t_1 and t_2 = $\int_{t_1}^{t_2} E(t)dt$

Also, $\int_0^\infty E(t)dt = 1$

- $F(t)$: Cumulative distribution function. Used for step experiments.

$$F(t) = \int_0^t E(t)\,dt$$, F(t) is the fraction of tracer/reactor contents exiting that has been in the reactor for less than time t

- t_m: Mean residence time

$$t_m = \frac{\int_0^\infty t\,E(t)\,dt}{\underbrace{\int_0^\infty E(t)\,dt}_{= \text{ to 1 by definition}}} = \int_0^\infty t\,E(t)\,dt$$

- σ^2: Variance. Provides an indication of the "spread" in the data

$$\sigma^2 = \int_0^\infty (t - t_m)^2\,E(t)\,dt$$